

NotesNPU

Private, offline AI lecture & meeting copilot - accelerated by the Snapdragon NPU on HP PCs

Snapdragon AI Lab Build & Present Challenge 2026 | Mohammed Haris Suhaib M |
<https://github.com/mdharishsuhaib/NotesNPU>

1. Overview

NotesNPU is a Windows desktop application for Snapdragon-powered HP PCs. It turns lectures, classes and meetings into timestamped transcripts, summaries, structured study notes, flashcards, quizzes and a grounded Q&A chatbot, completely on-device. Speech recognition runs Qualcomm AI Hub's Whisper model on the Snapdragon Hexagon NPU through the ONNX Runtime QNN Execution Provider. A local small language model (Phi-3.5-mini, INT4) produces the study material. No audio or text ever leaves the laptop, and it works without internet.

2. Problem

- Indian students attend 25-30 hours of lectures a week, often in fast, code-mixed Hindi-English, and struggle to take notes and listen at the same time.
- Cloud transcription tools (Otter, Fireflies, cloud Copilots) upload sensitive classroom and meeting audio to third-party servers.
- They need reliable internet and paid subscriptions, which are hard to get in hostels, tier-2/3 colleges, trains and field work.
- Running AI on the CPU drains the battery and heats the laptop, so it can't run through a whole day of classes.

3. Solution & key features

- Record or upload a lecture and get a timestamped transcript, fully on-device, even in airplane mode.
- Qualcomm AI Hub Whisper runs on the Hexagon NPU via ONNX Runtime QNN EP, leaving the CPU free and saving battery.
- A local LLM (Phi-3.5-mini INT4) generates a summary, structured notes, flashcards, a quiz and answers questions.
- Auto-extracts action items and deadlines ('exam next Monday', 'read chapter 6').
- Supports 10+ Indian languages, with Hindi lecture to English notes translation in the same pass.

Feature	What it does
Transcribe	Upload or record; pause-aware 30 s chunking; timestamps
Live mode	Mic streaming, text every ~8 s, on-device
Translate	Hindi / Tamil / Telugu / Bengali... to English
Study notes	Summary, key concepts, detailed notes, deadlines
Revise	Flashcards and 5-question MCQ quiz
Ask the lecture	Grounded Q&A chat on your transcript
Export	One-click Markdown notes
Benchmark	Built-in NPU vs CPU performance panel

4. AI models used

Task	Model	Source	Runtime / hardware
Speech-to-text (NPU)	Whisper-Base, precompiled QNN ONNX FP16	Qualcomm AI Hub	ONNX Runtime + QNN EP (HTP) on the Hexagon NPU
Speech-to-text (fallback)	Whisper-Base ONNX (INT8 decoder)	Hugging Face onnx-community	ONNX Runtime CPU EP
Study assistant	Phi-3.5-mini-instruct INT4 AWQ	Microsoft (Hugging Face)	onnxruntime-genai

5. Technical implementation

- Audio pipeline: 16 kHz mono resampling, pause-aware chunking (≤ 30 s windows cut at the quietest 100 ms), RMS silence gate to prevent Whisper hallucinations.
- NPU inference: AI Hub 'precompiled QNN ONNX' Whisper-Base for Snapdragon X Elite / X2 Elite (FP16 EPContext graphs, QAIRT 2.50). The encoder outputs cross-attention K/V caches directly. A fixed-shape single-step decoder (200-slot self-attention KV cache, additive attention mask, position ids) keeps every step on the HTP. The QNN plugin EP is registered at runtime (onnxruntime-qnn 2.x) with the HTP backend and burst performance mode.
- Decoding: forced prompt `<|sot|><|lang|><|task|><|notimestamps|>`, greedy decoding with timestamp suppression, n-gram repetition guard, automatic language detection, and translate-to-English mode for Indian languages.
- Study AI: onnxruntime-genai streaming generation with task-specific prompts, head+tail context fitting for long lectures, Markdown clean-up, and an instant extractive fallback (TF-IDF ranking, action-item detection, cloze MCQs).
- Robustness: automatic NPU to CPU to extractive fallback, so the same build runs on any Windows PC; built-in NPU vs CPU benchmark.
- UI: local Gradio web app (127.0.0.1 only) with Transcribe, Live mode, Study notes, Q&A and Benchmark tabs, plus Markdown export.

6. Why Snapdragon / NPU optimisation

- Whisper encoder = dense transformer over 1500 frames, a good fit for the Hexagon HTP's FP16 tensor engines (measured: 45.5 ms per 30 s of audio on X Elite).
- AI Hub precompiled context binaries: no on-device compile, instant start-up.
- Fixed-shape KV-cache decoder (200 slots) keeps every step on the NPU (3.8 ms/token).
- NPU does the speech, so the Oryon CPU stays free for the LLM and UI and live mode never stutters.
- Low-power NPU inference means a full day of lectures on battery, quiet and cool.

Device / backend	Speed	Encoder / 30 s	Decoder / token
Snapdragon X Elite, Oryon CPU (measured on Qualcomm AI Hub)	~14x real time (~6.1 s)	1286 ms	9.1 ms
Snapdragon X Elite, Hexagon NPU (measured on Qualcomm AI Hub)	~80x real time (~1.1 s)	45.5 ms	3.8 ms
NPU speed-up	~5.7x end to end	28.3x	2.4x

- Measured on a real Snapdragon X Elite (Qualcomm AI Hub hosted device): encoder 45.5 ms per 30 s of audio, decoder 3.8 ms per token.

- 100% of ops on the NPU: 556 encoder ops + 975 decoder ops on the Hexagon HTP, none on the CPU; peak memory 34 MB / 60 MB.
- Accuracy: NPU (FP16) encoder output matches the FP32 CPU reference with cosine similarity 0.999, and gives an identical, correct transcript of the sample lecture.
- The 85 s sample lecture takes ~1.1 s on the NPU vs ~6.1 s on the same Snapdragon X Elite's Oryon CPU.

7. Use cases & impact

Who	How NotesNPU helps
Students	Focus on understanding, not scribbling; revise with auto flashcards and quizzes.
Accessibility	Live captions for deaf and hard-of-hearing learners; notes for students with dyslexia or ADHD.
Regional language learners	Hindi / regional lectures turned into English notes, bridging the medium-of-instruction gap.
Professionals	Confidential meetings summarised with action items, and nothing is uploaded.
Teachers	Publish lecture notes and question banks from their own recordings.

8. Deployment & accessibility

- One-click setup.bat / run.bat: detects Snapdragon, installs native ARM64 deps, downloads AI Hub models.
- Runs as a local web app on 127.0.0.1, with offline mode enforced (HF_HUB_OFFLINE, no analytics).
- Automatic fallback: NPU to CPU to extractive engine, so it works on every Windows PC.
- Light install (-NoLLM, ~300 MB) for low-storage laptops.
- Open source (MIT) on GitHub, with CLI tools for transcription and benchmarking.

9. Roadmap

- LLM on the NPU: AI Hub Llama-3.2-3B / Phi-3.5 QNN binaries via Genie
- Speaker diarisation for meetings (who said what)
- Slide / whiteboard capture with on-device OCR, linked to the timeline
- Semantic search across a semester of lectures (on-device embeddings)
- MSIX installer on the Microsoft Store; HP AI Companion integration

10. Originality & licences

NotesNPU was designed and built from scratch by the participant during the Challenge Submission Period. It integrates open models from Qualcomm AI Hub (Whisper) and other open-source platforms (Phi-3.5-mini, ONNX Whisper) under their respective licenses. The application code is original and MIT-licensed.

Appendix: screenshot

NotesNPU

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1 Transcribe

2 Live mode

3 Study notes

4 Ask the lecture

5 NPU benchmark

Summary

Structured notes

Flashcards

Quiz

Key Concepts

- **Photosynthesis:** Process by which plants convert light energy into chemical energy (glucose) using CO_2 and H_2O .
- **Chloroplasts:** Organelles within plant cells where photosynthesis occurs, containing chlorophyll.
- **Light Dependent Reactions:** Stage involving conversion of light energy into chemical energy (ATP and NADPH).
- **Calvin Cycle:** Stages that utilize ATP and NADPH to synthesize glucose from CO_2 .
- **Factors Affecting Rate:** Light intensity, CO_2 concentration, and temperature influence photosynthesis efficiency.

Detailed Notes

- **Location of Photosynthesis** Mainly in leaf chloroplasts.
 - Occurs in Thylakoid Membranes during Light Dependent Reactions.
 - Splitting Water Molecules: Oxygen released as a byproduct.
 - Produces Energy Carriers: ATP & NADPH.
 - In Stroma during Calvin Cycle.
 - Fixation of CO_2 ; Formation of Glucose.
- **Importance of Photosynthesis**
 - Provides Oxygen: Essential for respiration.
 - Foundation of Food Chains: Primary source of organic matter.
- **Exam Preparation**
 - Draw and Label a Chloroplast: Understand its structure and function.
 - Explain Photosynthesis Significance: Discuss its role in ecosystems.
- Study Material